

Figure 15.13 An applied field causes domain boundaries to move. At saturation the sample has become a single domain, as on the right. Domain walls move easily in soft magnetic materials. Hard magnetic materials are alloyed to prevent their motion.

aligned against it. The domain wall separating them feels a force, the *Lorentz*[5] *force*, pushing it in a direction to make the favorably oriented domains grow at the expense of the others. As they grow, the magnetization of the material increases, moving up the *M*–*H* curve, finally saturating at M_s, when the whole sample is one domain oriented parallel to the field, as in Figure 15.13.

The saturation magnetization, then, is just the sum of all the atomic moments contained in a unit volume of material when they are all aligned in the same direction. If the magnetic dipole per atom is $n_m m_B$ (where n_m is the number of unpaired electrons per atom and m_B is the Bohr magneton), then the saturation magnetization is

$$M_s = \mu_o \frac{n_m m_B}{\Omega} \tag{15.8}$$

where Ω is the atomic volume. Iron has the highest because its atoms have the most unpaired electrons and they are packed close together. Anything that reduces either one reduces the saturation magnetization. Nickel and cobalt have lower saturation because their atomic moments are less; alloys of iron tend to be lower because the non-magnetic alloying atoms dilute the magnetic iron. Ferrites have much lower saturation both because of dilution by oxygen and because of the partial cancelation of atomic moments sketched in Figure 15.11.

There are, none the less, good reasons for alloying and making iron into ferrites. One relates to the coercive field, H_c. A good permanent magnet retains its magnetization even when placed in a reverse field, and this requires a high H_c. If domain walls move easily the material is a soft magnet with a low H_c; if the opposite, it is a hard one. To make a hard magnet it is necessary to pin the domain

[5] Hendrik Antoon Lorentz (1853–1928), Dutch mathematical physicist, friend and collaborator with Raleigh and Einstein, originator of key concepts in optics, electricity, relativity and hydraulics (he modeled and predicted the movement of water caused by the building of the Zuyderzee).

walls to stop them moving. Impurities, foreign inclusions, precipitates and porosity all interact with domain walls, tending to pin them in place (they act as obstacles to dislocation motion too, so magnetically hard materials are mechanically hard as well). And there are subtler barriers to domain-wall motion: *magnetic anisotropy* arising because certain directions of magnetization in the crystal are more favorable than others, and even the shape and size of the specimen itself—small, single-domain magnetic particles are particularly resistant to having their magnetism reversed.

The performance of a hard magnet is measured by its *energy product*, roughly proportional to the area of the magnetization loop. The higher the energy product, the more difficult it is to demagnetize it.

15.5 Materials selection for magnetic design

Selection of materials for their magnetic properties follows the new familiar pattern of translating design requirements followed by screening and ranking. In this case, however, the first screening step—'must be magnetic'—immediately isolates the special set of materials shown in the property charts of this chapter. The key selection issue is then whether a soft or hard magnetic response is required.

Soft magnetic devices

Electromagnets, transformers, electron lenses and the like have magnetic cores that must magnetize easily when the field switches on, be capable of creating a high flux density, yet lose their magnetism when the field is switched off. They do this by using soft, low H_c, magnetic materials, shown on the left of the chart of Figure 15.8. Soft magnetic materials are used to 'conduct' and focus magnetic flux. Figure 15.14 illustrates this. A current-carrying coil induces a magnetic field in the core. The field, instead of spreading in the way shown in Figure 15.3, is trapped in the extended ferro-magnetic circuit, as shown here. The higher the susceptibility χ, the greater is the magnetization and flux density induced by a given current loop, up to a limit set by its saturation magnetization M_s. The magnetic circuit conducts the flux from one coil to another, or from a coil to an

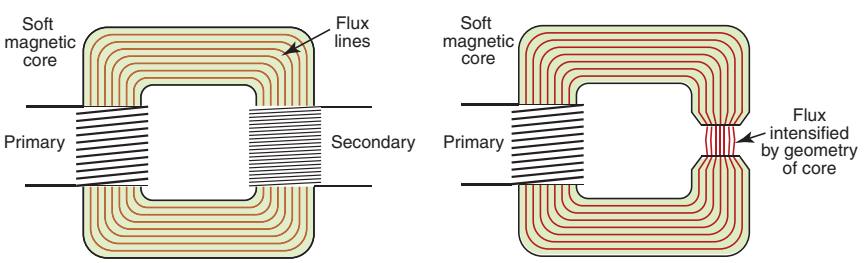


Figure 15.14 Soft magnets 'conduct' magnetic flux and are concentrated if the cross-section of the magnet is reduced.

air gap, where it is further enhanced by reducing the section of the core, where it is used to actuate or to focus an electron beam, as shown on the right of Figure 15.14. Thus, the first requirement of a good soft magnet is a high χ and a high M_s with a thin hysteresis loop like that sketched in Figure 15.15. The chart of Figure 15.9 shows these two properties. Permendur (Fe–Co–V alloys) and Metglass amorphous alloys are particularly good, but being expensive they are used only in small devices. Silicon–iron (Fe–1–4% Si) is much cheaper and easier to make in large sections; it is the staple material for large transformers.

Most soft magnets are used in AC devices, and then energy loss, proportional to the area of the hysteresis loop, becomes a consideration. Energy loss is of two types: the hysteresis loss already mentioned—it is proportional to switching frequency, f—and eddy current losses caused by the currents induced in the core itself by the AC field, proportional to f^2. In cores of transformers operating at low frequencies (meaning f up to 100 Hz or so) the hysteresis loss dominates, so for these we seek high M_s and a thin loop with a small area. Eddy-current losses are present but are suppressed when necessary by *laminating*: making the core from a stack of thin sheets interleaved with thin insulating layers that interrupt the eddy currents. At audio frequencies ($f < 50$ kHz) both types of loss become greater because they occur on every cycle and now there are more of them, requiring an even narrower loop and thinner laminates; here the more expensive Permalloys and Permendurs are used. At higher frequencies still (f in the MHz range) eddy-current loss, rising with f^2, becomes really serious and the only way to stop this is to use magnetic materials that are electrical insulators. That means ferrites (shown in yellow envelopes on both charts), even though they

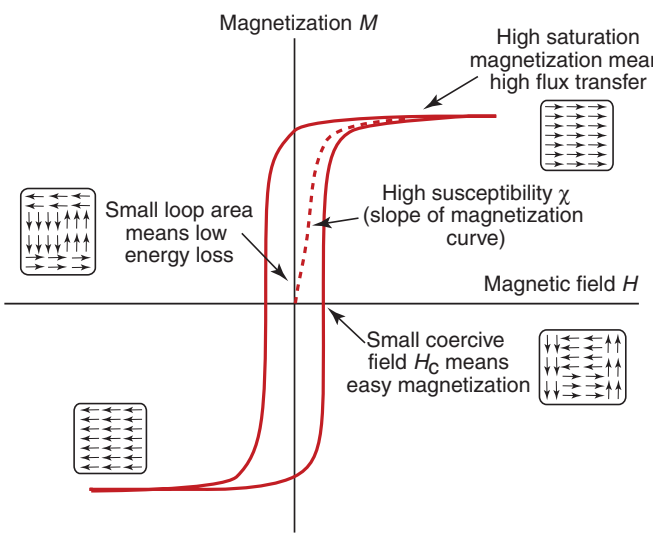


Figure 15.15 A characteristic hysteresis loop for a soft magnet.

Table 15.1 Materials for soft magnetic applications

Application	Frequency f	Material requirements	Material choice
Electromagnets	<1 Hz	High M_s, high χ	Silicon–iron Fe–Co alloys (Permendur)
Motors, low-frequency transformers	<100 Hz	High M_s, high χ	Silicon–iron AMA (amorphous Ni–Fe–P alloys)
Audio amplifiers, loudspeakers, microphones	<100 kHz	High M_s, very low H_c, high χ	Ni–Fe (Permalloy, Mumetal)
Microwave and UHF applications	<MHz	High M_s, very low H_c, electrical insulator	Cubic ferrites: MFe_2O_4 with M = Cu / Mn / Ni
Gigahertz devices	>100 MHz	Ultra low hysteresis, excellent insulator	Garnets

have a lower M_s. Above this ($f > 100$ MHz) only the most exotic ceramic magnets will work. Figure 15.15 summarizes the loop characteristics for soft magnets; Table 15.1 lists the choices, guided by the charts.

Hard magnetic devices

Many devices use permanent magnets: fridge doors, magnetic clutches, loudspeakers and earphones, DC motors, even frictionless bearings. For these, the key property is the remanence, M_R, since this is the maximum magnetization the material can offer without an imposed external field to keep it magnetized. High M_R, however, is not all. When a material is magnetized and the magnetizing field is removed, the field of the magnet itself tries to demagnetize it. It is this demagnetizing field that makes the material take up a domain structure that reduces or removes the net magnetization of the sample. A permanent magnet must be able to resist being demagnetized by it own field. A measure of its resistance is its coercive field, H_c, and this too must be large. A high M_R and a high H_c means a fat hysteresis loop like that of Figure 15.16. Ordinary iron and steel can become weakly magnetized with the irritating result that your screwdriver sticks to screws and your scissors pick up paper-clips when you wish they wouldn't, but neither are 'hard' in a magnetic sense. Here 'hard' means 'hard to demagnetize', and that is where high H_c comes in; it protects the magnet from itself.

Hard magnetic materials lie on the right of the $M_R - H_c$ chart of Figure 15.8. The workhorses of the permanent magnet business are the Alnicos and the hexagonal ferrites. An Alnico (there are many) is an alloy of aluminum, nickel, cobalt and iron—hence its name—with values of M_R as high as 10^6 A/m and a huge coercive field: 5×10^4 A/m. The hexagonal ferrites, too, have high M_R and H_c and, like the cubic soft-magnet ferrites, they are insulators. Being metallic, the Alnicos can be shaped by casting or by compacting and sintering powders; the ferrites can only be shaped by powder methods. Powders, of course, don't

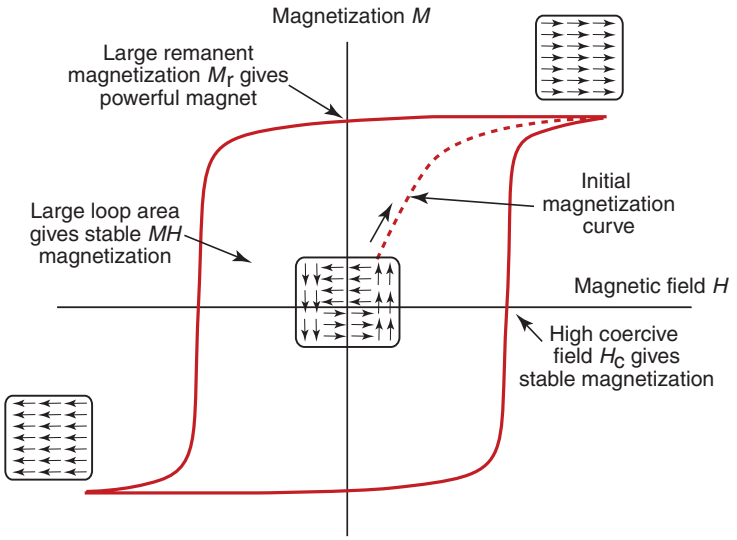


Figure 15.16 A characteristic hysteresis loop for a hard magnet.

have to be sintered—you can bond them together in other ways, and these offer opportunities. Magnetic particles mixed with a resin can be molded to complex shapes. Bonding them with an elastomer gives magnets that are flexible or squashy—the magnetic seal of a fridge door is an example.

The chart of Figure 15.8 shows three families of hard magnets with coercive fiels that surpass those of Alnicos and hexagonal ferrites. These, the products of research over the last 30 years, provide the intense permanent-magnet fields that have allowed the miniaturization of earphones, microphones and motors. The oldest of these is the precious metal Pt–Co family but they are so expensive that they have now all but disappeared. They are outperformed by the cobalt–samarium family based on the inter-metallic compound $SmCo_5$, and they, in turn, are outperformed by the recent development of the neodymium–iron–boron family. None of these are cheap, but in the applications they best suit, the quantity of material is sufficiently small and the product value sufficiently high that this is not an obstacle. Table 15.2 lists the choices, guided by the charts.

Magnetic storage of information

Magnetic information storage requires hard magnets, but ones with an unusual loop shape: one that is rectangular (called 'square'). The squareness means that a unit of the material can flip in magnetization from $+M_R$ to $-M_R$ and back when exposed to fields above $+H_c$ or $-H_c$. The unit thus stores one binary bit of information. The information is read by moving the magnetized unit past a read head, where it induces an electric pulse. The choice of the word 'unit' was deliberately vague because it can take several forms: discrete particles embedded in a polymer tape or disk, or as a thin magnetic film that has the ability to

Table 15.2 Materials for hard magnetic applications

Application	Material requirements	Material choice
Large permanent magnets	High M_R, high H_c and high energy product $(MH)_{max}$; low cost	Alnicos Hexagonal ferrites
Small high-performance magnets	Acoustic, electronic and miniature mechanical applications	Cobalt–samarium Platinum–cobalt Neodymium–boron
Information storage	Thin, 'square' hysteresis loop	Elongate particles of Fe_2O_3, Cr_2O_3 or hexagonal ferrite

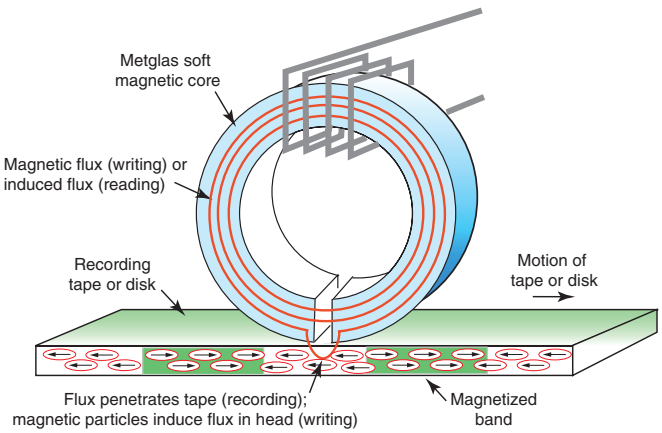


Figure 15.17 A magnetic read/write head. The gap in the soft magnetic head allows flux to escape, penetrating the tape and realigning the particles.

carry small stable domains called 'bubbles' that are magnetized in one direction, embedded in a background that is magnetized in another.

Figure 15.17 shows one sort of information storage system. A signal creates a magnetic flux in a soft magnetic core—often an amorphous metal alloy (AMA) because of its large susceptibility. The direction of the flux depends on the direction of the signal current; switching this reverses the direction of the flux. An air gap in the core allows field to escape, penetrating a tape or disk in which sub-micron particles of a hard magnetic material are embedded. When the signal is in one direction all particles in the band of tape or disk passing under the write-head are magnetized in one direction; when the signal is reversed, the direction of magnetization is reversed. The same head is used for reading. When the tape or disk is swept under the air gap, the stray field of the magnetized bands induces a flux in the soft iron core and a signal in the coil, which is amplified and processed to read it.